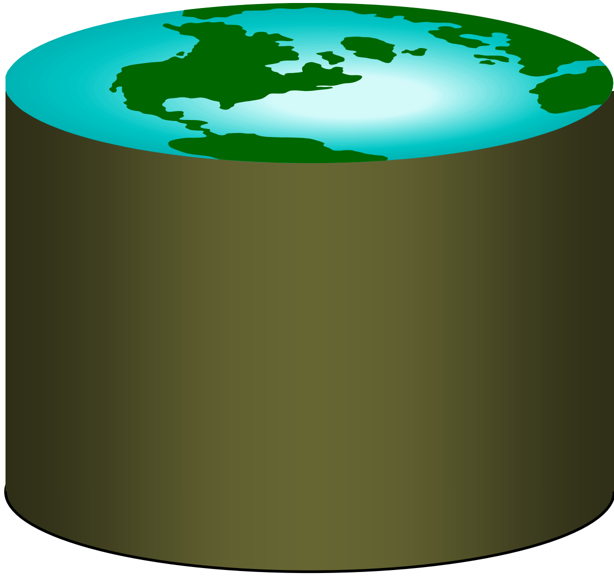
Chapter 4B: Tuple Relational Calculus



Relational Calculus

- **Comes in two flavors: *Tuple relational calculus* (TRC) and *Domain relational calculus* (DRC).**
- **Calculus has *variables, constants, comparison ops*, *logical connectives* and *quantifiers*.**
  - *TRC*: Variables range over (i.e., get bound to) *tuples*.
    - Like SQL.
  - *DRC*: Variables range over *domain elements* (= field values).
    - Like Query-By-Example (QBE)
  - Both TRC and DRC are simple subsets of first-order logic.
- **Expressions in the calculus are called *formulas*.**
- **Answer tuple is an assignment of constants to variables that make the formula evaluate to *true*.**

Tuple Relational Calculus

- ***Query*** **has the form: {*T* | *p*(*T*)}**
  - *p(T)* denotes a formula in which tuple variable *T* appears.
- ***Answer*** **is the set of all tuples *T* for which the *formula p*(*T*) evaluates to *true*.**
- ***Formula*** **is recursively defined:**
  - ❖start with simple *atomic formulas* (get tuples from relations or make comparisons of values)
  - ❖build bigger and better formulas using the *logical connectives*.

TRC Formulas

- *An Atomic formula is one of the following:*

  R $\in$ *Rel*

  R.a op S.b

  R.a op constant

  op is one of $<, >, =, \leq, \geq, \neq$

- **A *formula* can be:**
  - an atomic formula
  - $\neg p, p \wedge q, p \vee q$ where p and q are formulas
  - $\exists R(p(R))$ where variable *R* is a tuple variable
  - $\forall R(p(R))$ where variable *R* is a tuple variable

Free and Bound Variables

- **The use of quantifiers $\exists X$ and $\forall X$ in a formula is said to *bind* X in the formula.**
  - A variable that is not bound is free.
- **Let us revisit the definition of a query:**
  - $\{ T \mid p(T) \}$
- **There is an important restriction**
  - the variable T that appears to the left of `|' must be the *only* free variable in the formula $p(T)$.
  - in other words, all other tuple variables must be bound using a quantifier.

Selection and Projection

- **Find all sailors with rating above 7**

$$\{S \mid S \in Sailors \wedge S.rating > 7\}$$

  – Modify this query to answer: Find sailors who are older than 18 or have a rating under 9, and are called 'Bob'.

- **Find names and ages of sailors with rating above 7.**

$$\{S \mid \exists S1 \in Sailors(S1.rating > 7 \wedge S.sname = S1.sname \wedge S.age = S1.age)\}$$

Note, here *S* is a tuple variable of 2 fields (i.e. {S} is a projection of *sailors)*, since only 2 fields are ever mentioned and *S* is never used to range over any relations in the query.

Joins

Find sailors rated > 7 who've reserved boat #103

$$\{S \mid S \in Sailors \wedge S.rating > 7 \wedge \exists R(R \in Reserves \wedge R.sid = S.sid \wedge R.bid = 103)\}$$

Note the use of $\exists$ to find a tuple in Reserves that `joins with' the Sailors tuple under consideration.

Joins (continued)

Find sailors rated > 7 who've reserved boat #103

$$\{S \mid S \in Sailors \wedge S.rating > 7 \wedge$$
$$\exists R(R \in Reserves \wedge R.sid = S.sid$$
$$\wedge R.bid = 103)\}$$

Find sailors rated > 7 who've reserved a red boat

$$\{S \mid S \in Sailors \wedge S.rating > 7 \wedge$$
$$\exists R(R \in Reserves \wedge R.sid = S.sid$$
$$\wedge \exists B(B \in Boats \wedge B.bid = R.bid$$
$$\wedge B.color = \text{‘red’}))\}$$

- **Observe how the parentheses control the scope of each quantifier's binding. (Similar to SQL!)**

Division (makes more sense here???)

Find sailors who've reserved all boats

(hint, use $\forall$)

$$\{S \mid S \in Sailors \wedge \\ \forall B \in Boats\ (\exists R \in Reserves \\ (S.sid = R.sid \\ \wedge B.bid = R.bid))\}$$

- **Find all sailors *S* such that for each tuple *B* in Boats there is a tuple in Reserves showing that sailor *S* has reserved it.**

Division – a trickier example...

Find sailors who've reserved all Red boats

$$\{S \mid S \in Sailors \wedge \forall B \in Boats\ (B.color = \text{'red'} \Rightarrow \exists R(R \in Reserves \wedge S.sid = R.sid \wedge B.bid = R.bid))\}$$

Alternatively…

$$\{S \mid S \in Sailors \wedge \forall B \in Boats\ (B.color \neq \text{'red'} \vee \exists R(R \in Reserves \wedge S.sid = R.sid \wedge B.bid = R.bid))\}$$

$a \Rightarrow b$ is the same as $\neg a \vee b$

		b = T	b = F
a	T	T	F
	F	T	T

- If a is true, b must be true for the implication to be true. If a is true and b is false, the implication evaluates to false.
- If a is not true, we don't care about b, the expression is always true.

Unsafe Queries, Expressive Power

- **∃ syntactically correct calculus queries that have an infinite number of answers!** ***Unsafe*** **queries.**
  - e.g., $\left\{ S \mid \neg \left(S \in Sailors \right) \right\}$
  - Solution???? Don't do that!
- **Expressive Power (Theorem due to Codd):**
  - every query that can be expressed in relational algebra can be expressed as a safe query in DRC / TRC; the converse is also true.
- ***Relational Completeness*****: Query language (e.g., SQL) can express every query that is expressible in relational algebra/calculus. (actually, SQL is more powerful, as we will see...)**

Summary

- **The relational model has rigorously defined query languages — simple and powerful.**
- **Relational algebra is more operational**
  - useful as internal representation for query evaluation plans.
- **Relational calculus is non-operational**
  - users define queries in terms of what they want, not in terms of how to compute it. (*Declarative*)
- **Several ways of expressing a given query**
  - a *query optimizer* should choose the most efficient version.
- **Algebra and safe calculus have same *expressive power***
  - leads to the notion of *relational completeness.*

Addendum: Use of $\forall$

- $\forall x\ (P(x))$ - is only true if P(x) is true for *every* x in the universe
- Usually:

  $\forall \mathbf{x}\ ((\mathbf{x} \in \mathbf{Boats}) \Rightarrow (\mathbf{x.color} = \text{“}\mathbf{Red}\text{”}))$
- $\Rightarrow$ logical implication,

  a $\Rightarrow$ **b** means that if a is true, b must be true

  a $\Rightarrow$ **b** is the same as $\neg a \vee b$

Find sailors who've reserved all boats

$$\{S \mid S \in Sailors \wedge \forall B(\,(B \in Boats) \Rightarrow \exists R(R \in Reserves \wedge S.sid = R.sid \wedge B.bid = R.bid))\}$$

- **Find all sailors *S* such that for each tuple *B* either it is not a tuple in Boats or there is a tuple in Reserves showing that sailor *S* has reserved it.**

$$\{S \mid S \in Sailors \wedge \forall B(\neg(B \in Boats) \vee \exists R(R \in Reserves \wedge S.sid = R.sid \wedge B.bid = R.bid))\}$$

... reserved all red boats

$$\{S \mid S \in Sailors \wedge \\ \forall B(\ (B \in Boats \wedge B.color = \text{“red”}) \Rightarrow \\ \exists R(R \in Reserves \wedge S.sid = R.sid \\ \wedge B.bid = R.bid))\}$$

- **Find all sailors *S* such that for each tuple *B* either it is not a tuple in Boats or there is a tuple in Reserves showing that sailor *S* has reserved it.**

$$\{S \mid S \in Sailors \wedge \\ \forall B(\neg(B \in Boats) \vee (B.color \neq \text{“red”}) \vee \\ \exists R(R \in Reserves \wedge S.sid = R.sid \\ \wedge B.bid = R.bid))\}$$